

Password-Seeking with AMLD+0:

Singleton Enumeration, SAT, and CNF Pruning

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Abstract

This paper rewires singleton enumeration around the AMLD+0 architecture. The starting observation remains that a single designated mathematical query can hide an enormous search space. The AMLD+0 viewpoint adds a second distinction: a singleton *query* need not induce a singleton *settlement region*. A fixed conjecture T may be investigated by several verifier-gated settlement roles—proof, refutation, dependence, independence, and contradiction—while a distinguished Player 0 decides which enabled role receives the next service turn. Player 0 does not choose the selected role’s internal move and cannot alter verifier acceptance. Ordinary AMLD search is recovered when Player 0 replays or delegates to the declared baseline scheduler, so the present rewiring preserves the original search theory as a passive subcase.

Within this architecture, the Ocean, Universal Proof Horizon, password search, and structured-versus-opaque singleton distinction acquire a controller-level interpretation. Local navigators choose expansions inside channels; Player 0 allocates computational attention among channels; the shared verifier-governed BANK transmits accepted discoveries; and the verifier remains the fixed trust boundary. The password model then becomes a control case proving that scheduling can exploit information but cannot create information absent from the observation interface. The paper retains the original horizon, information-gap, CNF-pruning, execution-separation, and fair-isomorphism results, and adds AMLD+0 consequences concerning passive recovery, optional control, typed settlement horizons, and the separation of local navigation from global scheduling.

1 The AMLD+0 version of the central question

The original singleton-enumeration question begins with a one-element target set

$$\mathcal{T} = \{t\}, \quad |\mathcal{T}| = 1,$$

and observes that this says almost nothing about the number of legal states, legal moves, candidate histories, or search decisions surrounding t . That observation remains intact. AMLD+0 adds a second layer.

Suppose the mathematical object of study is one conjecture T . The query is singleton in the literal sense that the run is organized around one designated formula. But AMLD+0 does not force all successful computation to terminate in one undifferentiated state. It contains typed settlement roles

$$P, R, D, I, C,$$

sharing a verifier-governed BANK and a fixed verifier. A successful run may therefore terminate through one of several certificate-backed settlement channels, depending on the declared formal experiment.

This gives the first principle of the rewired paper.

Principle 1.1 (Singleton Query / Typed Settlement Principle). *A singleton mathematical query does not imply a singleton operational settlement region. In AMLD+0, one fixed target T may have several typed, verifier-gated modes of certified settlement, even though the object-level query itself remains unique.*

Thus at least three cardinalities must be kept separate:

1. the number of designated queries, which here is one;
2. the number of typed settlement regions admitted by the architecture;
3. the number of distinct certificates or histories capable of reaching any such region.

The complexity of a run may be large because of the third item, because of the second, or because the controller must decide how to allocate work among several search processes before any certificate is available.

2 AMLD+0 as the controller architecture

We use the foundational AMLD+0 separation between settlement-agent dynamics, shared verified memory, certification, and scheduling [5]. Fix a formal system F , admitted hypotheses Γ , a target T , a shared BANK B , a verifier V , a declared baseline scheduler σ_0 , and five settlement roles P, R, D, I, C .

Definition 2.1 (AMLD+0 singleton-settlement state). *A global AMLD+0 state is represented schematically by*

$$X = (B, S_P, S_R, S_D, S_I, S_C, H_0),$$

where B is the trusted BANK, each S_J is the internal search state of role $J \in \{P, R, D, I, C\}$, and H_0 is the controller-visible history and auxiliary state available to Player 0. The verifier V and baseline scheduler σ_0 are fixed components of the declared experiment rather than mutable coordinates of the ordinary run.

For a search-oriented channel one may have, for example,

$$S_J = (D_J, F_J, X_J, P_J, \Sigma_J, \dots),$$

with discovered states, frontier, expanded states, predecessor/certificate information, and heuristic or reconnaissance data. These are *local* coordinates. They belong to the settlement agent, not to Player 0.

Definition 2.2 (Player-Zero service event). *At a scheduling event, Player 0 selects*

$$J \in \{P, R, D, I, C, \varepsilon\}.$$

If J is an enabled settlement role, that role receives the next service turn and chooses its own legal internal action. If Player 0 selects the passive symbol ε , the next service decision is delegated to the declared baseline scheduler σ_0 . Player 0 does not select the chosen role’s internal move and cannot change verifier acceptance.

The distinction can be summarized as

$$\boxed{P_0 \text{ chooses the channel; } \quad J \text{ chooses the move; } \quad V \text{ judges the certificate.}}$$

This is the principal rewiring of the original higher-level search game. In the earlier presentation, the central search decision could be written

$$\text{Expand}(v), \quad v \in F.$$

In AMLD+0 that is ordinarily a *local agent action*. Player 0 acts one level above it by deciding which settlement process is serviced next.

3 The Ocean remains a singleton GBG inside AMLD+0

Consider a finite directed graph

$$G = (V, E)$$

with distinguished source s and target t . An edge $u \rightarrow v$ represents one legal inference step. As before, this is a one-player generalized board game in the sense of the GBG framework [2].

Definition 3.1 (Ocean GBG). *The Ocean generalized board game has game states V , initial state s , legal moves given by E , and singleton winning set*

$$\mathcal{W} = \{t\}.$$

A play is a directed path beginning at s , and a winning play reaches t .

Under unit edge cost the intrinsic winning horizon is

$$L^* = d_G(s, t).$$

Nothing about AMLD+0 changes this local graph fact. The important change is where the Ocean is placed in the architecture. An Ocean may be the local search geometry of one settlement role, while other roles explore different certificate spaces or metatheoretic obligations in parallel.

Thus the statement

$$L^* = d_G(s, t)$$

may describe the intrinsic horizon of one channel without describing the total AMLD+0 cost needed to discover, select, verify, and exploit that route.

Remark 3.2 (Local horizon versus controller cost). *A shortest route inside S_P is not automatically a shortest global AMLD+0 run. Global cost can include service given to other channels, verifier work, BANK operations, reconnaissance, controller computation, and any declared fallback schedule.*

Conversely, another channel may produce a certificate or BANK deposit that makes the globally cheapest settlement route different from the route initially visible to P .

4 Universal Proof Horizon inside a typed settlement architecture

The Universal Proof-Horizon viewpoint begins with an effective proof system, a mechanically checkable proof relation, and a computable proof-cost measure [1]. One may enumerate candidate proofs in increasing cost and return the first valid proof of the target. If a proof exists, this identifies a minimum-cost proof relative to the declared presentation and cost measure; if no proof exists, naive enumeration need not terminate.

For a fixed theorem T , let

$$\mathcal{W}_T = \{\pi : \pi \text{ verifies as a proof of } T\}.$$

The basic singleton lesson is still

$$\text{one desired theorem} \not\Leftarrow \text{small candidate-proof space.}$$

AMLD+0 refines this lesson. The Prover's proof horizon is only one possible route to a certified report. Write

$$\text{Sett}_J \subseteq \mathcal{X}$$

for the global AMLD+0 states in which the verifier has accepted a settlement certificate emitted through channel J . Without fixing a more detailed settlement taxonomy, define the positive settlement region by

$$\text{Sett}_{+0} := \bigcup_{J \in \{P, R, D, I, C\}} \text{Sett}_J.$$

When a directed accepted-state metric is declared, one may define channel horizons

$$H_J(X) = d(X, \text{Sett}_J)$$

and the intrinsic typed-settlement horizon

$$H_{+0}(X) = d(X, \text{Sett}_{+0}) = \min_J H_J(X),$$

whenever these distances are finite and measured in the same declared unit. This quantity is geometric: it describes the cheapest accepted route in the full state graph. It does not imply that Player 0 knows which J attains the minimum.

5 Passive rewiring: the old paper is a subcase

The AMLD+0 architecture was designed so that ordinary AMLD is recovered when Player 0 makes no adaptive scheduling intervention [5]. Specializing that fact to singleton search yields the following.

Theorem 5.1 (Singleton Passive-Rewiring Theorem). *Let $\mathcal{A}$ be an ordinary AMLD run on a fixed*

singleton query T , with settlement roles P, R, D, I, C , shared BANK B , verifier V , and baseline scheduler σ_0 . There is an AMLD+0 extension $\mathcal{A}^{+0}$ such that the projection onto the ordinary AMLD coordinates reproduces the run of $\mathcal{A}$ whenever Player 0 either

1. selects the same enabled role that σ_0 would select, or
2. selects ε and delegates the service decision to σ_0 .

For stochastic or nondeterministic agents, exact trajectory recovery is understood relative to the same fixed random or choice trace.

Proof. Adjoin Player 0 without changing any local agent transition rule, BANK update rule, verifier rule, or initial ordinary AMLD state. Induct on service events. If the projected ordinary states agree immediately before event n , then scheduler replay or silent delegation selects the same role as σ_0 . That role receives the same local state and, relative to the same random or choice trace when necessary, makes the same internal move. The same verifier judges the same proposed deposits or settlement certificates. Hence the projected state agrees after event n . The induction holds for every finite prefix. $\square$

Corollary 5.2 (Conservative reinterpretation of the original paper). *Every result in the original singleton-enumeration analysis that depends only on the ordinary search trajectory remains available as the passive AMLD+0 baseline. Active Player-Zero control is therefore an additional experimental variable rather than a replacement of the earlier theory.*

This is important methodologically. The AMLD+0 rewrite does not obtain new results by silently changing the verifier, proof system, or local search rules. It adds a controller coordinate whose effect can be compared directly with a declared baseline.

6 The higher-level game is now a scheduling game

A local search role may still maintain

$$S_J = (D_J, F_J, X_J, P_J, \Sigma_J, \dots)$$

and may still choose an action such as

$$\text{Expand}_J(v), \quad v \in F_J.$$

The controller-level game, however, has actions of the form

$$\text{Serve}(J) \quad \text{or} \quad \varepsilon.$$

This separates two optimization problems that the original paper treated in a single navigation layer:

local navigation: which legal action should channel J take?

global scheduling: which enabled channel should receive the next service turn?

The separation matters because a locally excellent navigator can be globally starved, while a globally intelligent scheduler cannot repair a channel whose local policy ignores all useful structure.

Principle 6.1 (Two-Level Navigation Principle). *AMLD+0 search quality is jointly controlled by local settlement-agent policies and the Player-Zero service policy. Neither layer in general determines the other.*

7 The shared BANK changes the geometry

The BANK makes the several search channels more than an unrelated portfolio. Verifier-authorized intermediate results can become shared resources. If accepted deposits are support-safe and conservative relative to the declared formal system, they can change the *explicit* search landscape without silently enlarging theoremhood.

Suppose channel J deposits a verified lemma λ into B . Another channel K may then obtain a shorter local route because λ is now an available trusted premise. In controller language, service allocated to J can therefore have option value even when J never produces the final settlement certificate.

This leads to an AMLD+0 version of the Ocean metaphor: Player 0 is not merely choosing among five independent boats. It is allocating expeditions whose verified discoveries can redraw the shared navigational chart.

Remark 7.1 (No verifier shortcut). *A BANK deposit changes what is explicitly available to search only after the verifier authorizes it under the declared rules. Player 0 cannot promote an unverified formula to trusted status merely because doing so would shorten an estimated horizon.*

8 Password space as an AMLD+0 control case

Password search is useful because it isolates a case where a singleton target is embedded in a large candidate space that is almost empty in navigational signal. Let $\mathcal{A}$ be an alphabet of size k , let passwords have length m , and let

$$\Omega = \mathcal{A}^m.$$

Let the hidden accepted password be $W \in \Omega$.

A rule-complete full state may be represented by

$$x = (W, G, U, R),$$

where G is the set of tested candidates, $U = \Omega \setminus G$, and R is the response transcript. The searcher observes only

$$O(W, G, U, R) = (G, U, R).$$

A legal move is $\text{Guess}(q)$ for $q \in U$.

Definition 8.1 (Observation-admissible policy). *A policy is O-admissible when its action depends only on its own randomness and the history of available observations and actions, not on hidden coordinates of the full state.*

The point of importing this model into AMLD+0 is not that password guessing literally requires the five metalogical channels. Rather, it provides a clean theorem about the limits of scheduling: duplicating or rescheduling searchers cannot manufacture a directional signal that the interface never reveals.

Theorem 8.2 (Player-Zero Noncreation of Information). *Let W be uniformly distributed on a candidate set Ω of size N . Suppose every enabled worker receives only equality feedback—“correct” or “incorrect”—for complete guesses, and the federation enforces global nonrepetition: once any candidate has been tested, no worker tests it again. Player 0 observes only the combined legal transcript and worker/controller history and has no side channel, hidden advice, or access to W . Then any AMLD+0-style schedule together with any collection of such workers induces, conditional on all controller and worker randomness, an ordering of the N candidates. Consequently the rank of W in the total sequence of distinct tested candidates is uniform on $\{1, \dots, N\}$.*

Proof. Fix all controller and worker randomness. Until success, every legal response is identical: “incorrect.” Thus the entire deterministic federation, including Player 0’s scheduling decisions, chooses a first untested candidate, then a second, and so on. Because global nonrepetition is enforced, these choices form an ordering of Ω . Uniform W has uniform rank in any fixed ordering. Averaging over the previously fixed randomness does not change the rank distribution. $\square$

Corollary 8.3 (Scheduling can exploit information but cannot create it). *In the equality-only opaque password model, adding Player 0, adding more workers, or adaptively reallocating service cannot improve the information content of the observation stream. Any improvement must come from legitimate prior information, a richer observation map, additional legal queries, or a change in the problem itself.*

This is the sharp conceptual boundary for active AMLD+0. Scheduling is a control resource, not an oracle.

9 Structured and opaque singleton search

The password control case motivates the same distinction as the original paper, now stated relative to both local policies and Player 0.

Definition 9.1 (Structured singleton search). *A singleton search problem is structured when the observations available to the search architecture contain features correlated with accessibility, distance, expected settlement cost, or other legitimate progress information toward certified settlement.*

Definition 9.2 (Opaque singleton search). *A singleton search problem is opaque when candidates that remain untested are indistinguishable with respect to all information legally available to the search architecture, except through direct testing.*

Principle 9.3 (AMLD+0 Singleton Navigation Principle). *A singleton query does not prevent intelligent navigation or intelligent scheduling. What matters is whether the observation history contains information correlated with progress and whether the local agents and Player 0 can exploit that information without crossing the verifier trust boundary.*

This creates a useful four-way distinction:

1. the query may be singleton;
2. the intrinsic certificate space may be huge;
3. the observation geometry may be structured or opaque;
4. the controller may allocate work well or poorly relative to the structure that is actually observable.

10 DATA MIND under AMLD+0

The original paper described DATA MIND abstractly as choosing

$$v \in F$$

so as to maximize expected progress toward the unique target. Under AMLD+0, that description belongs naturally to a *local* settlement agent or to a navigation module used by such an agent.

For example, a local value estimate may be written

$$Q_J(v) \approx \text{estimated value of expanding } v \text{ inside channel } J.$$

Player 0, by contrast, may maintain a service value

$$Q_0(J \mid H_0) \approx \text{estimated value of giving the next service turn to } J.$$

These are different objects. The first ranks actions within a channel. The second ranks channels given the current shared history.

A clean experimental decomposition is therefore

$$\boxed{\text{local candidate ordering} \longrightarrow \text{verifier/BANK updates} \longrightarrow \text{Player-Zero service allocation}}$$

with feedback in the opposite direction through the newly observed shared state.

Neither layer changes the formal verifier. Heuristics change the order in which legal work is attempted; the verifier alone determines whether a proposed certificate or BANK addition becomes trusted.

11 Player-Zero control value and the passive baseline

Because ε delegates to the baseline scheduler, active control can be compared against a precisely defined passive run. Let

$$C_{\sigma_0}(X)$$

be expected declared settlement cost from state X under the baseline scheduler, and let $C_{\pi_0}(X)$ be the corresponding cost under a Player-Zero policy π_0 , with the same local agents, verifier, BANK rules, and cost accounting.

Define the expected controller gain

$$G_0(X; \pi_0) := C_{\sigma_0}(X) - C_{\pi_0}(X).$$

Positive G_0 means the intervention lowers expected declared work; negative G_0 means it raises expected declared work relative to baseline.

Proposition 11.1 (Optional-Control Dominance). *Assume that Player 0's policy class contains the passive policy that selects ε at every scheduling event, and that using this passive action introduces no extra charged cost beyond the baseline mechanism it invokes. Then*

$$\inf_{\pi_0} C_{\pi_0}(X) \leq C_{\sigma_0}(X).$$

Proof. The always- ε policy belongs to the admissible Player-Zero policy class and reproduces the baseline cost by hypothesis. The infimum over a set containing that policy cannot exceed its value. $\square$

Remark 11.2. *This proposition is an option-value statement, not a speedup theorem. An arbitrary active Player-Zero policy may be worse than baseline. The theorem says only that a controller optimization problem with costless fallback does not have a worse best achievable value than the fallback it contains.*

12 Information-constrained Bellman gap for opaque passwords

Let Ω contain $N \geq 1$ candidate passwords, let W be uniformly distributed on Ω , and use the equality-only observation model above. Let Π_O denote the class of policies that are O -admissible, nonrepeating on failed queries, and continue until success. For $\pi \in \Pi_O$, let T_π be its number of guesses. For every full state the intrinsic full-information Bellman cost is one guess.

Theorem 12.1 (Information-Constrained Bellman Gap). *For the equality-only password game, every admissible possibly randomized nonrepeating policy π satisfies, for $0 \leq k \leq N$,*

$$\Pr(T_\pi \leq k) = \frac{k}{N}$$

and

$$\mathbb{E}[T_\pi] = \frac{N+1}{2}.$$

If

$$V_{\text{Bayes}}^{\text{id}, \mu} = 1$$

is the full-observation expected settlement value and

$$V_{\text{Bayes}}^{O, \mu} := \inf_{\pi \in \Pi_O} \mathbb{E}[T_\pi],$$

then

$$\boxed{\frac{V_{\text{Bayes}}^{O, \mu}}{V_{\text{Bayes}}^{\text{id}, \mu}} = \frac{N+1}{2}}.$$

For $N = 2^n$ the information-constrained Bellman gap is exponential in n .

[Theorem retained from the Atlas-assisted expansion of the original paper; presentation adapted here to the AMLD+0 viewpoint.]

Proof. Condition on the internal randomness of the policy. Until success every observed answer is “incorrect,” so a deterministic nonrepeating realization specifies an ordering $q_1, \dots, q_N$ of the candidates. Since W is uniform, its rank in that ordering is uniform on $\{1, \dots, N\}$. Therefore

$$\Pr(T_\pi \leq k) = \frac{k}{N}, \quad \mathbb{E}[T_\pi] = \frac{1}{N} \sum_{j=1}^N j = \frac{N+1}{2}.$$

Averaging over policy randomness leaves these identities unchanged. A full-information controller guesses W immediately, so the stated ratio follows. $\square$

From the AMLD+0 viewpoint, the theorem applies to the federation as a whole whenever the combined controller/worker observation is still equality-only. The controller can alter service order but not the information-theoretic rank symmetry.

13 Opaque singleton search under scheduling

Theorem 13.1 (Opaque Singleton No-Polynomial-Settlement). *Let W be a uniformly random n -bit password and suppose the only feedback from a guess is equality with W . If an O -admissible target-blind policy makes at most $q(n)$ distinct guesses, then*

$$\Pr(\text{settlement within } q(n) \text{ guesses}) \leq \min \left\{ 1, \frac{q(n)}{2^n} \right\}.$$

Equality holds when the policy tests exactly $\min\{q(n), 2^n\}$ distinct candidates unless success occurs earlier. Consequently, for every polynomial q ,

$$\Pr(\text{settlement within } q(n) \text{ guesses}) \rightarrow 0.$$

Thus the uniform opaque-password family has no high-probability polynomial settlement policy in the observation-constrained equality-query model.

[Theorem retained from the Atlas-assisted expansion of the original paper; presentation adapted here to the AMLD+0 viewpoint.]

Proof. A policy using at most $q(n)$ guesses tests at most $\min\{q(n), 2^n\}$ distinct candidates. Conditional on any fixed realization of its randomness, those candidates occupy at most that many positions in the induced ordering. Uniform W lies among them with probability at most

$$\min \left\{ 1, \frac{q(n)}{2^n} \right\}.$$

If the policy continues through exactly that many distinct candidates unless success occurs earlier, the bound is attained. For polynomial q , $q(n)/2^n \rightarrow 0$. $\square$

Corollary 13.2 (AMLD+0 opaque-control corollary). *If every AMLD+0 worker and Player 0 are restricted to the same equality-only transcript and all charged candidate tests are counted globally, then merely splitting the work among channels or workers does not evade the theorem. A polynomial global test budget still covers only polynomially many members of a 2^n -element candidate set.*

This gives a strong negative control for controller experiments. If active Player 0 performs substantially better on a singleton benchmark, the gain must be traced to legitimate structure, prior information, richer observations, shared certified discoveries, or a change in the allowed action set—not to scheduling in the abstract.

14 Constraint revelation in CNF and 3-CNF

Password space becomes more informative when compared with SAT. A known CNF formula exposes constraints before a complete assignment is tested. Partial assignments may falsify clauses and prune the prefix tree.

Fix a variable ordering $x_1, \dots, x_n$ and let φ be a satisfiable CNF formula. Let $P_k(\varphi)$ be the assignments to $x_1, \dots, x_k$ that have not yet falsified any clause, let

$$W_k := |P_k(\varphi)|,$$

and define

$$J_k := \log_2 \left(\frac{2^k}{W_k} \right) = k - \log_2 W_k.$$

Theorem 14.1 (CNF Prefix-Pruning Monotonicity). *If φ is satisfiable, then*

$$W_0 = 1, \quad 1 \leq W_k \leq 2^k, \quad W_{k+1} \leq 2W_k,$$

and

$$W_n = \# \text{SAT}(\varphi).$$

Consequently

$$\boxed{0 = J_0 \leq J_1 \leq \dots \leq J_n = n - \log_2 \# \text{SAT}(\varphi).}$$

If the satisfying assignment is unique, $J_n = n$. The statement applies in particular to satisfiable and uniquely satisfiable 3-CNF formulas.

[Theorem retained from the Atlas-assisted expansion of the original paper; presentation adapted here to the AMLD+0 viewpoint.]

Proof. The empty assignment is the unique length-zero prefix, so $W_0 = 1$. Restricting any satisfying assignment to the first k variables gives a surviving prefix, so $W_k \geq 1$, while trivially $W_k \leq 2^k$. Each surviving k -prefix has at most two extensions, hence $W_{k+1} \leq 2W_k$. At depth n , a complete

assignment survives exactly when it satisfies φ , so $W_n = \# \text{SAT}(\varphi)$. Finally,

$$J_{k+1} - J_k = 1 - \log_2 \left(\frac{W_{k+1}}{W_k} \right) \geq 0.$$

The endpoint formula and uniquely satisfiable case follow. $\square$

For AMLD+0, J_k is an example of a legitimate observable feature that may inform either local action choice, global service allocation, or both. The theorem does not imply that SAT becomes easy; it proves only that SAT may reveal progressively asymmetric information that the ideal password oracle withholds.

15 Four equivalent local horizons

The Ocean construction, GBGs, the Universal Proof-Horizon viewpoint, and Bellman settlement geometry all contain a notion of distance to success. For a unit-cost one-player proof game they coincide.

Theorem 15.1 (Singleton GBG–Proof-Horizon–Bellman Equivalence). *Let $G = (V, E)$ be an effectively generated, locally finite, directed one-player GBG with initial state s and singleton winning set $\{t\}$. Suppose each legal move has unit cost and the reachable GBG transition system is step-for-step identical to the associated sequential proof system. If t is reachable from s , let L^* be the minimum winning match length, $h(t)$ the minimum number of inference steps in a proof of t , $V(s)$ the Bellman settlement value, and $d_G(s, t)$ the directed graph distance. Then*

$$\boxed{L^* = h(t) = V(s) = d_G(s, t).}$$

[Theorem retained from the Atlas-assisted expansion of the original paper; presentation adapted here to the AMLD+0 viewpoint.]

Proof. The step-for-step identity makes legal matches from s to t exactly the admissible sequential proof paths. Thus minimum match length equals minimum proof length, and both equal directed graph distance. Unit action costs make the cost of a settling trajectory equal to its edge count, so the Bellman value equals the same distance. $\square$

Remark 15.2 (AMLD+0 interpretation). *The equality is a local intrinsic invariant. It can identify the horizon of a particular proof channel or encoded proof game. It does not collapse the controller-level quantities*

$$H_{+0}(X), \quad C_{\pi_0}(X), \quad C_{\sigma_0}(X)$$

into the same number. Typed alternatives, BANK effects, service allocation, and discovery overhead remain additional coordinates.

16 Shortest certificates do not control AMLD+0 execution cost

Definition 16.1 (Irrelevant expansion). *In a singleton-target directed search graph with target t , an expansion of v is irrelevant if*

$$d_G(v, t) = \infty,$$

so no continuation from v can contribute to a certificate reaching t .

Theorem 16.2 (Horizon–Execution Separation). *For every $L \geq 1$ and $M \geq 1$, there exists a finite singleton-target one-player GBG with an exact search policy π such that*

$$L^* = V(s) = L,$$

the returned certificate has exactly L legal moves, but π performs at least M irrelevant state expansions before completing it. Thus certificate stretch

$$\frac{\text{returned certificate length}}{L^*} = 1$$

does not bound search overhead.

[Theorem retained from the Atlas-assisted expansion of the original paper; presentation adapted here to the AMLD+0 viewpoint.]

Proof. Construct a unique successful path

$$s = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_L = t$$

and at least M reachable decoy states that do not lead to t . Give every edge unit cost. The shortest successful path has length L . Let π expand the M decoys first and then the successful path. Its predecessor map still reconstructs an optimal L -edge certificate, while at least M irrelevant expansions were performed. Since M is arbitrary, certificate stretch alone cannot bound overhead. $\square$

AMLD+0 adds a second separation on top of this theorem. Even if every local channel eventually returns an optimal certificate relative to its own graph, Player 0 may allocate arbitrarily large service to unproductive channels before servicing the winning route. Conversely, a good Player-Zero policy can reduce wasted cross-channel work without changing the length of the final certificate. Therefore at least the following should be reported separately:

- final certificate cost or length;
- local expansion counts by channel;
- verifier and BANK costs;
- Player-Zero service counts by channel;
- total charged settlement cost;
- baseline-versus-active controller difference.

17 Efficient representations and fair settlement degree

Theorem 17.1 (Efficient GBG Isomorphism Preserves Fair Settlement Degree). *Consider two families of finitely encoded, exact singleton-target GBGs, $\mathcal{G}$ and $\mathcal{H}$, equipped with settlement systems M_G and M_H . Suppose there are mutually inverse translations f and g , computable with polynomial declared primitive work, between encoded instances and corresponding state/action representations such that initial states and settlement status are preserved, legal transitions are preserved bijectively, action costs are preserved, encoded sizes are polynomially related, and terminal certificates/verifier acceptance are transportable with polynomial declared primitive work in both directions. Then the associated settlement pairs lie in the same fair settlement degree, and for corresponding states x and $f(x)$,*

$$\boxed{V_{M_G}(x) = V_{M_H}(f(x)).}$$

[Theorem retained from the Atlas-assisted expansion of the original paper; presentation adapted here to the AMLD+0 viewpoint.]

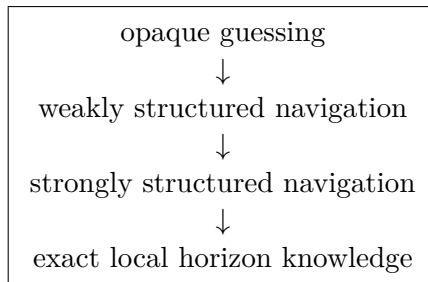
Proof. Each settling trajectory in $\mathcal{G}$ transports through f to a trajectory in $\mathcal{H}$ with the same accumulated action cost, and g provides the converse. Taking infima gives equality of Bellman values. The assumed polynomial construction, size, answer, and certificate-transport bounds provide fair reductions in both directions, hence equality of fair settlement degree [4]. $\square$

To preserve an AMLD+0 experiment rather than merely a local GBG, an even stronger recoding is appropriate: the translation should also preserve typed channel identity, controller observations, legal Player-Zero service actions, BANK/verifier semantics, and the charged costs of controller decisions. A map that preserves only reachability can hide precisely the scheduling geometry that AMLD+0 was introduced to study.

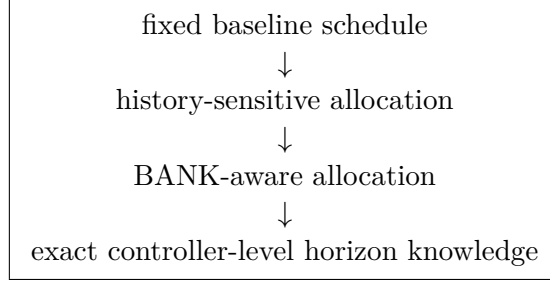
18 A hierarchy of AMLD+0 singleton-search regimes

The original paper placed singleton search on a spectrum from opaque guessing to exact shortest-path knowledge. AMLD+0 keeps that local spectrum and adds a controller dimension.

At the local level:



At the controller level:



These spectra are independent in principle. One may have a highly intelligent local prover under a poor scheduler, or a sophisticated Player 0 allocating turns among locally weak agents. The most capable architecture combines informative local state, effective local policies, useful shared verified memory, and a controller that exploits only legitimate observations.

19 The rewired hierarchy of quantities

The original singleton-search synthesis separated target cardinality, intrinsic horizon, observable information geometry, navigation policy, and actual cost. AMLD+0 requires one additional layer and makes the distinction between local and global policy explicit:



No arrow is reversible in general. In particular:

- one query does not imply one settlement mode;
- a short intrinsic horizon does not imply that an admissible policy can identify it cheaply;
- informative local geometry does not imply good scheduling;
- good scheduling does not imply informative local geometry;
- an optimal final certificate does not imply low total search cost;
- no controller policy is permitted to alter verifier acceptance.

This layered non-equivalence is the conceptual space in which AMLD+0 is meant to operate.

20 Experimental consequences

The rewiring suggests a direct experimental protocol for any singleton benchmark.

First run the declared passive baseline σ_0 . Record the settlement report, certificate, channel service counts, per-channel expansions, verifier work, BANK activity, and total charged cost. Then run an active Player-Zero policy with the same formal system, initial BANK, local agents, verifier, cost model, and contamination firewall. Compare the runs using quantities such as

$$G_0 = C_{\sigma_0} - C_{\pi_0}.$$

For a structured Ocean, one can ask whether Player 0 learns to allocate service toward channels whose verified discoveries shorten subsequent horizons. For an opaque password control, the Player-Zero Noncreation of Information theorem predicts no information-theoretic gain from scheduling alone. For SAT or other constraint-rich domains, features such as prefix pruning can be tested as legitimate controller observations.

This produces a sharper empirical question than “did the intelligent system solve the singleton target faster?”:

Which part of the gain came from local navigation, which came from shared verified discoveries, which came from Player-Zero allocation, and which was already present in the passive baseline?

That decomposition is precisely what the AMLD+0 architecture makes measurable.

21 Conclusion

Singleton enumeration remains a useful way to expose a basic misconception in search: a one-element target set need not be computationally small. The AMLD+0 viewpoint strengthens the lesson by separating the uniqueness of the mathematical query from the multiplicity of certified settlement routes and from the controller problem of deciding where computation should go next.

The Ocean remains a valid one-player singleton GBG at the local level. The Universal Proof Horizon remains an intrinsic measure of shortest certified routes. Password space remains the opaque control case. CNF pruning remains an example of visible structure. What changes is the architectural level at which navigation is interpreted.

A local settlement agent may decide which vertex, lemma, assignment, or certificate candidate to inspect. Player 0 decides which enabled settlement process receives the next turn. The shared BANK transmits only verifier-authorized resources. The verifier remains the trust boundary. Passive Player 0 recovers the old AMLD trajectory; active Player 0 exposes a new, separately measurable source of operational performance.

The central conclusion of the rewired paper is therefore:

singleton target complexity is not only a geometry of candidates;
under AMLD+0 it is also a geometry of certified attention allocation.

But that additional control power has a precise limit. Player 0 may exploit structure already present in legal observations and verified shared state; it cannot schedule information into existence, and it cannot schedule around the verifier.

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